

Complete Heat Transfer Analysis: Cylinder in Square Enclosure with All Radiation Paths

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1 Problem Statement

Analyze heat loss from a steel cylinder enclosed in a plywood box, including **all** convective and radiative heat transfer paths:

Parameter	Symbol	Value
Cylinder diameter	D	0.25 m
Cylinder temperature	T_{cyl}	90°C (363 K)
Minimum air gap	δ	0.10 m
Inner box side	w_{in}	0.45 m
Plywood thickness	t	0.01 m
Outer box side	w_{out}	0.47 m
Ambient temperature	T_{∞}	20°C (293 K)
Analysis length	L	0.30 m
Steel emissivity (oxidized)	ε_1	0.80
Plywood emissivity	ε_2	0.90

2 Complete Thermal Network

The complete thermal network includes **five** resistances with two parallel radiation paths:

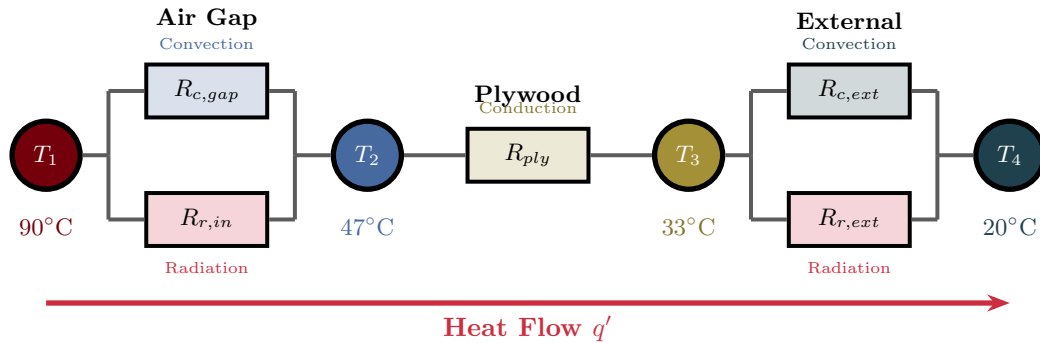


Figure 1: Complete thermal resistance network with parallel convection and radiation at both the air gap and external surfaces.

3 Surface Areas and View Factors

Per unit length:

$$A_1 = \pi D = \pi \times 0.25 = 0.785 \text{ m}^2/\text{m} \quad (\text{cylinder}) \quad (1)$$

$$A_2 = 4w_{in} = 4 \times 0.45 = 1.80 \text{ m}^2/\text{m} \quad (\text{inner box}) \quad (2)$$

$$A_3 = 4w_{out} = 4 \times 0.47 = 1.88 \text{ m}^2/\text{m} \quad (\text{outer box}) \quad (3)$$

View factors:

- $F_{1 \rightarrow 2} = 1$ (cylinder completely enclosed by inner box)
- $F_{3 \rightarrow \text{surroundings}} = 1$ (outer box is convex, radiates to large surroundings)

4 Resistance Calculations

4.1 Air Gap Convection ($R_{c,gap}$)

Shape factor for cylinder in square enclosure:

$$S = \frac{2\pi}{\ln(1.08 \cdot w_{in}/D)} = \frac{2\pi}{\ln(1.944)} = 9.45 \text{ m}^{-1} \quad (4)$$

Rayleigh and Nusselt numbers (with $\Delta T = 43^\circ\text{C}$):

$$Ra = \frac{g\beta\Delta T\delta^3}{\nu\alpha} = 2.7 \times 10^6 \quad (5)$$

$$Nu = 0.046 \cdot Ra^{1/3} = 6.4 \quad (6)$$

$$k_{eff} = k_{air} \cdot Nu = 0.0283 \times 6.4 = 0.181 \text{ W}/(\text{m} \cdot \text{K}) \quad (7)$$

$$R_{c,gap} = \frac{1}{S \cdot k_{eff}} = \frac{1}{9.45 \times 0.181} = \boxed{0.583 \text{ (m} \cdot \text{K)/W}} \quad (8)$$

4.2 Inner Radiation ($R_{r,in}$): Cylinder $\rightarrow$ Inner Plywood

For enclosed radiation between cylinder and square box:

$$\frac{1}{R_{r,in}} = \frac{\sigma(T_1^4 - T_2^4)}{T_1 - T_2} \cdot \frac{1}{\frac{1 - \varepsilon_1}{\varepsilon_1 A_1} + \frac{1}{A_1} + \frac{1 - \varepsilon_2}{\varepsilon_2 A_2}} \quad (9)$$

Denominator terms:

$$\frac{1 - \varepsilon_1}{\varepsilon_1 A_1} = \frac{0.20}{0.80 \times 0.785} = 0.318 \quad (10)$$

$$\frac{1}{A_1} = 1.273 \quad (11)$$

$$\frac{1 - \varepsilon_2}{\varepsilon_2 A_2} = \frac{0.10}{0.90 \times 1.80} = 0.062 \quad (12)$$

Linearized radiation coefficient: $h_{r,in} = 7.0 \text{ W}/(\text{m}^2 \cdot \text{K})$

$$R_{r,in} = \frac{1}{h_{r,in} \cdot A_1} = \frac{1}{7.0 \times 0.785} = \boxed{0.182 \text{ (m} \cdot \text{K)/W}} \quad (13)$$

4.3 Combined Air Gap Resistance

$$R_{gap} = \left(\frac{1}{R_{c,gap}} + \frac{1}{R_{r,in}} \right)^{-1} = \left(\frac{1}{0.583} + \frac{1}{0.182} \right)^{-1} = \boxed{0.139 \text{ (m} \cdot \text{K)/W}} \quad (14)$$

4.4 Plywood Conduction (R_{ply})

$$R_{ply} = \frac{t}{k_{ply} \cdot A_2} = \frac{0.01}{0.12 \times 1.80} = \boxed{0.046 \text{ (m} \cdot \text{K)/W}} \quad (15)$$

4.5 External Convection ($R_{c,ext}$)

$$R_{c,ext} = \frac{1}{h_{conv} \cdot A_3} = \frac{1}{7.0 \times 1.88} = \boxed{0.076 \text{ (m} \cdot \text{K)/W}} \quad (16)$$

4.6 External Radiation ($R_{r,ext}$): Outer Plywood $\rightarrow$ Surroundings

For a convex surface radiating to large surroundings:

$$q_{rad} = \varepsilon_2 \sigma A_3 (T_3^4 - T_\infty^4) \quad (17)$$

Linearized coefficient (at $T_3 \approx 33^\circ\text{C}$):

$$h_{r,ext} = \varepsilon_2 \sigma (T_3^2 + T_\infty^2)(T_3 + T_\infty) = 0.9 \times 5.67 \times 10^{-8} \times (306^2 + 293^2)(306 + 293) = \boxed{5.5 \text{ W/(m}^2 \cdot \text{K)}} \quad (18)$$

$$R_{r,ext} = \frac{1}{h_{r,ext} \cdot A_3} = \frac{1}{5.5 \times 1.88} = \boxed{0.097 \text{ (m} \cdot \text{K)/W}} \quad (19)$$

4.7 Combined External Resistance

$$R_{ext} = \left(\frac{1}{R_{c,ext}} + \frac{1}{R_{r,ext}} \right)^{-1} = \left(\frac{1}{0.076} + \frac{1}{0.097} \right)^{-1} = \boxed{0.043 \text{ (m} \cdot \text{K)/W}} \quad (20)$$

4.8 Total Resistance

$$R_{total} = R_{gap} + R_{ply} + R_{ext} = 0.139 + 0.046 + 0.043 = \boxed{0.228 \text{ (m} \cdot \text{K)/W}} \quad (21)$$

5 Heat Loss Results

$$q' = \frac{T_{cyl} - T_\infty}{R_{total}} = \frac{90 - 20}{0.228} = \boxed{307 \text{ W/m}} \quad (22)$$

$$Q = q' \times L = 307 \times 0.30 = \boxed{92 \text{ W}} \quad (23)$$

6 Temperature Distribution

$$T_1 = 90^\circ\text{C} \quad (\text{cylinder}) \quad (24)$$

$$T_2 = T_1 - q' R_{gap} = 90 - 307 \times 0.139 = \boxed{47^\circ\text{C}} \quad (\text{inner plywood}) \quad (25)$$

$$T_3 = T_2 - q' R_{ply} = 47 - 307 \times 0.046 = \boxed{33^\circ\text{C}} \quad (\text{outer plywood}) \quad (26)$$

$$T_4 = 20^\circ\text{C} \quad (\text{ambient}) \quad (27)$$

7 Model Comparison

Table 1: Effect of including radiation at each location

Model	R_{total} [(m·K)/W]	Q (30 cm) [W]	vs Base
1. Convection only	0.705	29.8	(base)
2. + Inner radiation	0.261	80.5	+170%
3. + External radiation (Complete)	0.228	92.3	+210%

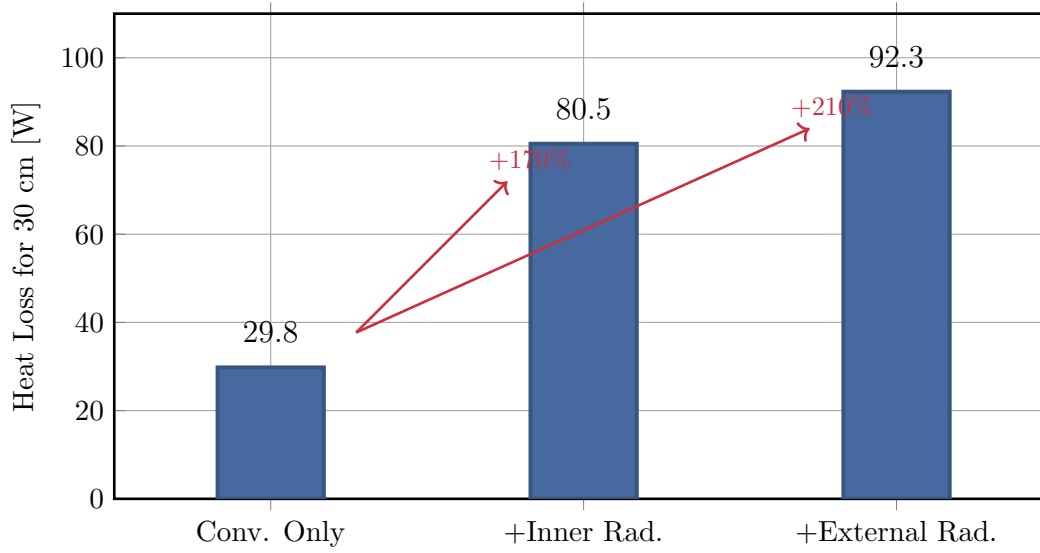


Figure 2: Heat loss increases significantly with each radiation path added.

8 Resistance Breakdown

Table 2: Thermal resistance breakdown for complete model

Component	Resistance [(m·K)/W]	Fraction	ΔT [°C]	Effect of Radiation
<i>Air Gap</i>				
Convection only	0.583	—	—	—
Radiation only	0.182	—	—	—
Combined	0.139	61%	43	−76%
Plywood (conduction)	0.046	20%	14	—
<i>External</i>				
Convection only	0.076	—	—	—
Radiation only	0.097	—	—	—
Combined	0.043	19%	13	−44%
TOTAL	0.228	100%	70	

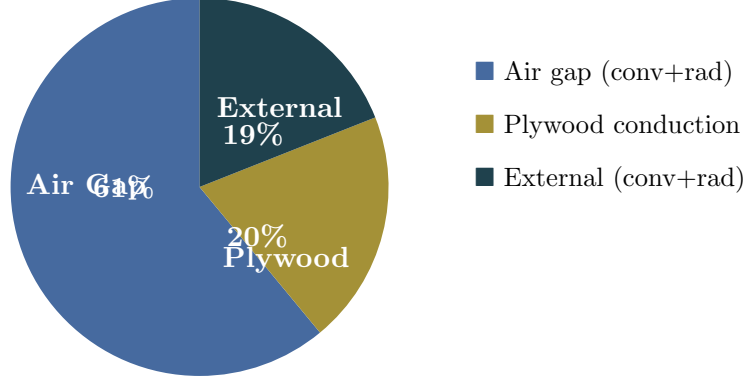


Figure 3: Resistance distribution in the complete model.

9 Key Findings

1. **Both radiation paths significantly increase heat loss:**
 - Inner radiation (cylinder $\rightarrow$ inner plywood): +170%
 - External radiation (outer plywood $\rightarrow$ surroundings): additional +15%
 - **Total increase:** +210% (from 30 W to 92 W)
2. **Inner radiation is dominant:** Reduces air gap resistance by 76% (from 0.58 to 0.14).
3. **External radiation is also significant:** Reduces external resistance by 44% (from 0.076 to 0.043). The external radiation coefficient $h_{r,ext} = 5.5 \text{ W}/(\text{m}^2 \cdot \text{K})$ is comparable to convection $h_{conv} = 7 \text{ W}/(\text{m}^2 \cdot \text{K})$.
4. **Final temperature profile:** $90^\circ\text{C} \rightarrow 47^\circ\text{C} \rightarrow 33^\circ\text{C} \rightarrow 20^\circ\text{C}$
5. **Plywood provides 20% of total resistance:** The main insulation comes from the air gap (61%), even with radiation present.

10 Conclusions

Table 3: Final summary of heat loss for 30 cm cylinder length

Heat Transfer Model	Heat Loss [W]
Convection only (incomplete)	30
With inner radiation only (incomplete)	81
Complete model (all paths)	92

The complete analysis shows that **radiation accounts for 68% of total heat loss** (62 W out of 92 W). Ignoring either radiation path leads to significant underestimation:

- Ignoring all radiation: underestimates by 68%
- Ignoring external radiation only: underestimates by 13%

For design purposes: To reduce heat loss, focus on reducing radiation, especially at the inner surface (low-emissivity coating on the cylinder or reflective barrier).